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OPTOELECTRONIC STRUCTURE

Abstract

An optoelectronic structure is provided. The optoelectronic structure includes an optical device; and a housing covering the optical device. The optical device is configured to transmit at least one optical signal to an outside of the housing in different directions.

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Background/Summary

BACKGROUND
1. Field of the Disclosure
[0001] The instant disclosure relates to, amongst other things, an optoelectronic structure.
2. Description of Related Art
[0002] In current vehicular LiDAR (Light Detection and Ranging) systems, there are generally

mechanical, solid-state, and fully solid-state types, with miniaturization primarily focused on solid-state and fully solid-state systems. Solid-state LiDAR, also known as “solid-state optical radar,” incorporates minimal moving parts. It utilizes MEMS (Micro-Electro-Mechanical Systems) mechanisms for a compact, electronic design that integrates the traditionally larger mechanical structures onto silicon-based chips through microelectronic processes. This design enables vertical one-dimensional scanning through MEMS mirrors, with the entire device achieving horizontal scanning by rotating 360 degrees. On the other hand, fully solid-state LiDAR systems employ the Optical Phased Array (OPA) approach, which consists of an array of multiple light sources. By controlling the phase timing differences of each light source's emission, a main light beam with a specific direction is synthesized. This main beam can then be controlled to scan in various directions, achieving millimeter-level radar accuracy. This advancement aligns with the future trends of LiDAR technology towards solid-state, miniaturization, and cost reduction.

[0003] However, in the design of purely solid-state LiDAR systems, an external prism design is still used to achieve scanning in different directions, indicating that there remains room for further miniaturization in LiDAR systems.

SUMMARY

[0004] According to one example embodiment of the instant disclosure, an optoelectronic structure includes an optical module; and a housing covering the optical module. The optical module is configured to provide at least two optical signals, and wherein the at least two optical signals pass through the housing in different directions.

[0005] According to another example embodiment of the instant disclosure, an optoelectronic structure includes an optical emission module and an optical transceiver module. The optical emission module is configured to emit a first optical signal along a first path. The optical transceiver module is configured to receive the first optical signal and emit a second optical signal along a second path. The second path overlaps the first path in a first direction which is substantially perpendicular to the first path or the second path.

[0006] According to another example embodiment of the instant disclosure, an optoelectronic structure includes a housing having a first side and a second side different from the first side; and an optical module covered by the housing. The optical module is configured to emit and/or receive a light passing through the first side of the housing and configured to emit and/or receive another light passing through the second side of the housing.

[0007] In order to further understanding of the instant disclosure, the following embodiments are provided along with illustrations to facilitate appreciation of the instant disclosure; however, the appended drawings are merely provided for reference and illustration, and do not limit the scope of the instant disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1**A is a schematic top view of an optoelectronic structure in accordance with an embodiment of the instant disclosure.

[0009] FIG. **1**B illustrates a schematic cross-sectional view along line A-A in FIG. **1**A.

[0010] FIG. **2** is a schematic top view of an optoelectronic structure in accordance with an embodiment of the instant disclosure.

[0011] FIG. **3** is a schematic cross-sectional view of an optoelectronic structure in accordance with an embodiment of the instant disclosure.

[0012] FIG. **4** is a schematic cross-sectional view of an optoelectronic structure in accordance with an embodiment of the instant disclosure.

[0013] FIG. **5**A, FIG. **5**B, FIG. **5**C, FIG. **5**D, FIG. **5**E, FIG. **5**F, FIG. **5**G, FIG. **5**H, FIG. **5**I, FIG. **5**J and FIG. **5**K illustrate one or more stages of an example of a method for manufacturing an optoelectronic structure in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0014] The following disclosure provides for many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to explain certain aspects of the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed or disposed in direct contact, and may also include embodiments in which additional features are formed or disposed between the first and second features, such that the first and second features are not in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] As used herein, spatially relative terms, such as “beneath,” “below,” “above,” “over,” “on,” “upper,” “lower,” “left,” “right,” “vertical,” “horizontal,” “side” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

[0016] Present disclosure provides an innovative structure for the protective cover of optical systems, featuring openings that allows laser light signals to pass through. To achieve a broader scanning range and more accurate distance measurement, it also integrates a light guiding component, such as a prism, into the carrier equipped with the optical system. This integration facilitates scanning and distance measuring in various directions, enhancing the system's overall performance and versatility.

[0017] FIG. **1**A is a schematic top view of an optoelectronic structure **1** in accordance with an embodiment of the instant disclosure. FIG. **1**B illustrates a schematic cross-sectional view along line A-Ain FIG. **1**A. As shown in FIG. **1**A and FIG. **1**B, the optoelectronic structure **1** may include a substrate **10**, a photonic component **121**, an electronic component **123**, a photonic component **131**, an electronic component **132** and a laser component **133**. The substrate **10** may be or include a carrier. The substrate **10** may be composed of a silicon-based material, leveraging the well-established properties of silicon to enhance the device's performance and integration capabilities. Silicon-based substrates are chosen for their excellent electrical, thermal, and mechanical properties, making them ideal for a wide range of applications. The substrate **10** may include an interconnection structure, which may include such as a plurality of conductive traces and/or a plurality of conductive vias. The interconnection structure may include a redistribution layer (RDL) **103**. The redistribution layer **103** may be disposed on an upper surface **101** of the substrate **10**.

[0018] Referring to FIG. **1**A and FIG. **1**B, the photonic component **121** may be disposed over the substrate **10** and the electronic component **123** may be stacked on the photonic component **121**. In some embodiments of the present disclosure, the photonic component **121** includes photonic integrated circuits (PIC). The photonic integrated circuits may be used for emitting laser pulses, guiding the optical path, and receiving the light reflected back. Moreover, the PIC of the photonic component **121** may include a waveguide **1215**. The waveguide **1215** is configured to direct light waves from one point to another within the chip. By precisely controlling the shape and material properties of the waveguides, light's propagation path can be effectively managed, facilitating functions such as routing, splitting, combining, and modulating optical signals. As shown in FIG. **1***i*, the photonic component **121** may include a surface **1210** (e.g., an upper surface) facing away from the substrate **10**. Further, the surface **1210** may include an active surface of the photonic component **121**. In addition, at least one electrical connection **125** may be configured to electrically connect the active surface of the photonic component **121** to the redistribution layer **103** of the substrate **10**. Thus, the photonic component **121** and the substrate **10** may be electrically connected to each other through the electrical connection **125**. In some embodiments of the present disclosure, the electrical connection **125** includes a conductive wire.

[0019] In some embodiments of the present disclosure, the electronic component **123** includes electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **1**B, the electronic component **123** may include a surface **1230** (e.g., a lower surface) facing the surface **1210** of the photonic component **121**. Further, the surface **1230** may include an active surface of the electronic component **123**. That is, the active surface of the photonic component **121** may face the active surface of the electronic component **123**. Moreover, a plurality of electrical connections **126** is arranged between the surface **1210** of the photonic component **121** and the surface **1230** of the electronic component **123** and configured to electrically connect the active surface of the photonic component **121** to the active surface of the electronic component **123**. Thus, the photonic component **121** and the electronic component **123** may be electrically connected to each other through the electrical connections **126**. In some embodiments of the present disclosure, the electrical connection **126** includes a micro-bump.

[0020] The photonic component **121** and the electronic component **123** may jointly be considered as an optical transceiver module **12**. The optical transceiver module **12** may be electrically connected to the substrate **10** through the at least one electrical connection **125**.

[0021] The photonic component **131** may be disposed over the substrate **10** and the electronic component **132** and the laser component **133** may be stacked on the photonic component **131**. In some embodiments of the present disclosure, the photonic component **131** includes photonic integrated circuits (PIC). The photonic integrated circuits may be used for emitting laser pulses, guiding the optical path, and receiving the light reflected back. Moreover, the PIC of the photonic component **131** may include a waveguide **1315**. The waveguide **1315** is configured to direct light waves from one point to another within the chip. By precisely controlling the shape and material properties of the waveguides, light's propagation path can be effectively managed, facilitating functions such as routing, splitting, combining, and modulating optical signals. As shown in FIG. **1***i*, the photonic component **131** may include a surface **1310** (e.g., an upper surface) facing away from the substrate **10**. Further, the surface **1310** may include an active surface of the photonic component **131**. In addition, at least one electrical connection **135** may be configured to electrically connect the active surface of the photonic component **131** to the redistribution layer **103** of the substrate **10**. Thus, the photonic component **131** and the substrate **10** may be electrically connected to each other through the electrical connection **135**. In some embodiments of the present disclosure, the electrical connection **135** includes a conductive wire.

[0022] In some embodiments of the present disclosure, the electronic component **132** includes electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **1**B, the electronic component **132** may include a surface **1320** (e.g., a lower surface) facing the surface **1310** of the photonic component **131**. Further, the surface **1320** may include an active surface of the electronic component **132**. That is, the active surface of the photonic component **131** may face the active surface of the electronic component **132**.

[0023] The laser component **133** may include a distributed feedback laser. The distributed feedback laser may be used for emitting highly coherent and wavelength-precise light beams. As shown in FIG. **1**B, the laser component **133** may include a surface **1330** (e.g., a lower surface) facing the surface **1310** of the photonic component **131**. Further, the surface **1330** may include an active surface of the laser component **133**. That is, the active surface of the photonic component **131** may face the active surface of the laser component **133**.

[0024] Referring to FIG. **1**B, the surface **1320** of the electronic component **132** and the surface **1330** of the laser component **133** may be in abutment with the surface **1310** of the photonic component **131**. The photonic component **131**, the electronic component **132** and the laser component **133** may be configured in a chip-to-chip assembly. The photonic component **131**, the electronic component **132** and the laser component **133** may be electrically connected to each other.

[0025] The photonic component **131**, the electronic component **132** and the laser component **133** may jointly be considered as an optical emission module **13**. The optical emission module **13** may be electrically connected to the substrate **10** through the at least one electrical connection **135**.

[0026] Moreover, since the optical transceiver module **12** may be electrically connected to the substrate **10** through the electrical connection **125** and the optical emission module **13** may be electrically connected to the substrate through the electrical connection **135**, the optical transceiver module **12** and the optical emission module **13** may be electrically coupled to each other through the substrate **10**.

[0027] As shown in FIG. **1**A and FIG. **1**B, the optoelectronic structure **1** may further include lens structures **141** and **142** and an optical guiding element **15**. The lens structure **141** may be disposed between the optical transceiver module **12** and the optical emission module **13**. In some embodiments of the present disclosure, the lens structure **141** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **141** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. Moreover, the lens structure **142** is disposed between the optical transceiver module **12** and the optical guiding element **15**. In some embodiments of the present disclosure, the lens structure **142** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **142** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. In some embodiments of the present disclosure, the optical guiding element **15** includes a beam splitter which is used to divide an incoming light beam into two parts.

[0028] In some embodiments of the present disclosure, the optoelectronic structure **1** may further include an electronic component **16** disposed on the substrate **10**. In some embodiments of the present disclosure, the electronic component **16** includes an integrated passive device (IPD).

[0029] Further, referring to FIG. **1**A and FIG. **1**B, the optoelectronic structure **1** may include a housing **11**. The housing **11** may be disposed on the upper surface **101** of the substrate **10**. The housing **11** may cover the optical transceiver module **12** (which may include the photonic component **121** and the electronic component **123**), the optical emission module **13** (which may include the photonic component **131**, the electronic component **132** and the laser component **133**), the lens structures **141** and **142** and the optical guiding element **15**. That is, the housing **11** is configured to protect the optical transceiver module **12** and the optical emission module **13**. The housing **11** may include a lateral portion **114** and an upper portion **115** adjacent to the lateral portion **114**. In some embodiments of the lateral portion **114** is located at a lateral side of the housing **11**. That is, the lateral portion **114** of the housing **11** may be in abutment with the upper surface **101** of the substrate **10**. In some embodiments of the present disclosure, the upper portion **115** is located at an upper side of the housing **11**. That is, the upper portion **115** may be spaced apart from the upper surface **101** of the substrate **10**. The lateral portion **114** of the housing **11** may include an opening **111**. As shown in FIG. **1**B, the opening **111** may be substantially aligned with the optical guiding element **15** in a direction which may be substantially parallel to the upper surface **101** of the substrate **10**. The upper portion **115** of the housing **11** may include an opening

112. As shown in FIG. **1**B, the opening **112** may be disposed directly above the optical guiding element **15** and substantially aligned with the optical guiding element **15** in a direction which may be substantially perpendicular to the upper surface **101** of the substrate **10**.

[0030] Referring to FIG. **1**B, the housing **11** may include a heat sink **110** at its upper side. In some embodiments of the present disclosure, the housing **11** may include a thermal conductive material. The electronic component **123** may be connected to the upper side of the housing **11**. A surface **1231** (e.g., an upper surface) of the electronic component **123** may be attached to the upper side of the housing **11** through a thermal interface material (TIM) **127**. In some embodiments of the present disclosure, the thermal interface material **127** includes a thermal paste. That is, the housing **11** with the heat sink **110** and the thermal interface material **127** may provide a thermal dissipation path for the optical transceiver module **12**. Moreover, the electronic component **132** and the laser component **133** may be connected to the upper side of the housing **11**. A surface **1321** (e.g., an upper surface) of the electronic component **132** and a surface **1331** (e.g., an upper surface) of the laser component **133** may be attached to the upper side of the housing **11** through a thermal interface material (TIM) **137**. In some embodiments of the present disclosure, the thermal interface material **137** includes a thermal paste. That is, the housing **11** with the heat sink **110** and the thermal interface material **137** may provide a thermal dissipation path for the optical emission module **13**. In addition, because the upper surfaces of the electronic component **123**, the electronic component **132** and the laser component **133** may be all attached to the upper side of the housing **11**, it can be understood that the upper surfaces of the electronic component **123**, the electronic component **132** and the laser component **133** may be substantially coplanar with each other.

[0031] As shown in FIG. **1**A and FIG. **1**B, the optical emission module **13** may emit an optical signal L**1** to the optical transceiver module **12** along a direction X**1**. In some embodiments of the present disclosure, the optical signal L**1** includes at least one laser light beam. In some embodiments of the present disclosure, the waveguide **1315** of the photonic component **131** is configured to transmit the optical signal L**1**. The optical signal L**1** emitted from the optical emission module **13** may pass through the lens structure **141**. Therefore, it can be known that the path of the optical signal L**1** may be substantially along the direction X**1** and may pass through the lens structure **141**.

[0032] The optical transceiver module **12** is configured to receive and/or detect the optical signal L**1** emitted from the optical emission module **13**. Then, the optical transceiver module **12** may emit an optical signal L**2** to the optical guiding element **15** along a direction X**2**. In some embodiments of the present disclosure, the optical signal L**2** includes at least one laser light beam. In some embodiments of the present disclosure, the waveguide **1215** of the photonic component **121** is configured to transmit the optical signal L**2**. The optical signal L**2** emitted from the optical transceiver module **12** may pass through the lens structure **142**. Therefore, it can be known that the path of the optical signal L**2** may be substantially along the direction X**2** and may pass through the lens structure **142**. In some embodiments of the present disclosure, the direction X**1** and the direction X**2** are substantially parallel to each other. Thus, the path of the optical signal L**1** may be substantially parallel to the path of the optical signal L**2**. Moreover, referring to FIG. **1**A, the path of the optical signal L**1** may overlap the path of the optical signal L**2** in a direction Y**1**, which may be substantially perpendicular to the path of the optical signal L**1** or the path of the optical signal L**2**. Referring to FIG. **1**A, the direction X**1** and the direction X**2** are opposite to each other. In addition, the photonic component **121** of the optical transceiver module **12** may receive the optical signal L**1** and emit the optical signal L**2** at the same lateral side **128**.

[0033] The optical guiding element **15** may divide the optical signal L**2** emitted from the optical transceiver module **12** into two parts. That is, after the optical guiding element **15** receives the optical signal L**2** emitted from the optical transceiver module **12**, the optical guiding element **15** may generate an optical signal L**3** along a direction X**3** and an optical signal L**4** along a direction Z**1**. The direction X**3** may be substantially identical to, or parallel to the direction X**2**. The direction

Z**1** may be substantially perpendicular to the direction X**3** or the direction X**2**. Thus, the optical guiding element **15** may split the optical signal L**2**, emitted by the optical transceiver module **12**, into two optical signals, L**3** and L**4**. Optical signal L**3** is guided in the direction X**3**, which may be substantially the same as the direction X**2** of optical signal L**2**, while optical signal L**4** is directed along the direction Z**1**, differing from the direction X**2** of optical signal L**2**.

[0034] The optical signal L**3** generated from the optical guiding element **15** may advance towards the opening **111** of the housing along the direction X**3**, thereby passing through the opening **111** of the housing **11** and emitting towards an exterior of the housing **11** of the optoelectronic structure **1**. Therefore, it can be known that the path of the optical signal L**3** may be substantially along the direction X**3** and pass through the opening **111** of the housing **11**.

[0035] The optical signal L**4** generated from the optical guiding element **15** may advance towards the opening **112** of the housing along the direction Z**1**, thereby passing through the opening **112** of the housing **11** and emitting towards an exterior of the housing **11** of the optoelectronic structure **1**. Therefore, it can be known that the path of the optical signal L**4** may be substantially along the direction Y**4** and pass through the opening **111** of the housing **11**.

[0036] Given the above, it can be known that the optoelectronic structure **1** may provide at least two optical signals in at least two different directions. Likewise, the optoelectronic structure **1** may be capable of receiving at least two optical signals, each reflected from external objects along at least two distinct directions. The external object may include a movable objects and/or an immovable objects, such as vehicle and building.

[0037] Referring to FIG. **1**A and FIG. **1**B, an external optical signal L**5**, reflected off an external object, may pass through the opening **111** of the housing **11** and enter an interior of the housing **11** along a direction X**5**. The optical transceiver module **12** may receive the optical signal L**5** in the direction X**5**. In some embodiments of the present disclosure, the waveguide **1215** of the photonic component **121** is configured to receive the optical signal L**5**. After the optical transceiver module **12** receive the optical signal L**5**, the optical transceiver module **12** may transmit information related to the optical signal L**5** to a processor/controller through the substrate **10**. Further, an external optical signal L**6**, reflected off an external object, may pass through the opening **112** of the housing **11** and enter an interior of the housing **11** along a direction Z**2**. The optical guiding element may receive the optical signal L**6** in the direction Z**2** and redirect the optical signal L**6** to advance toward the optical transceiver module **12**. In some embodiments of the present disclosure, the waveguide **1215** of the photonic component **121** is configured to receive the optical signal L**6**. After the optical transceiver module **12** receive the optical signal L**6**, the optical transceiver module **12** may transmit information related to the optical signal L**6** to the processor/controller through the substrate **10**.

[0038] FIG. **2** is a schematic top view of an optoelectronic structure **2** in accordance with an embodiment of the instant disclosure. As shown in FIG. **2**, the optoelectronic structure **2** may include a substrate **20**, a housing **21**, a photonic component **221**, an electronic component **223**, a photonic component **231**, an electronic component **232** and a laser component **233**. The substrate **20** is the same as, or similar to, the substrate **10** shown in FIG. **1**A and FIG. **1**B. The photonic component **231** may be disposed on and electrically connected to the substrate **20**, and the electronic component **232** and the laser component **233** may be stacked on the photonic component **231**. The photonic component **231** is the same as, or similar to, the photonic component **131** shown in FIG. **1**A and FIG. **1**B. The electronic component **232** is the same as, or similar to, the electronic component **132** shown in FIG. **1**A and FIG. **1**B. The laser component **233** is the same as, or similar to, the laser component **133** shown in FIG. **1**A and FIG. **1**B. The photonic component **231**, the electronic component **232** and the laser component **233** may be electrically connected to each other and thus may jointly be considered as an optical emission module **23** of the optoelectronic structure **2**.

[0039] The photonic component **221** may be disposed on and electrically connected to the substrate

20 and the electronic component **223** may be stacked on the photonic component **221**. The photonic component **221** is similar to the photonic component **121** shown in FIG. **1**A and FIG. **1***i*, with the distinction that the photonic component **221** may emit at least two optical signals. Moreover, the electronic component **223** is the same as, or similar to, the electronic component **123** shown in FIG. **1**A and FIG. **1**B. The photonic component **221** and the electronic component **223** may be electrically connected to each other and thus may jointly be considered as an optical transceiver module **22** of the optoelectronic structure **2**.

[0040] The optoelectronic structure **2** may further include lens structures **241**, **242** and **243** and an optical guiding element **25**. The lens structure **241** may be disposed on the substrate **20** and between the photonic component **221** and the photonic component **231**. That is, the lens structure **241** may be located between the optical transceiver module **22** and the optical emission module **23** of the optoelectronic structure **2**. The lens structure **241** is the same as, or similar to, the lens structure **141** shown in FIG. **1**A and FIG. **1**B. The lens structures **242** and **243** may be disposed on the substrate **20**. The lens structure **242** is the same as, or similar to, the lens structure **142** shown in FIG. **1**A and FIG. **1**B. The lens structure **242** is configured to allow one of the at least two optical signals emitted from the photonic component **221** to pass through. Moreover, the lens structure **243** is the same as, or similar to, the lens structure **142** shown in FIG. **1**A and FIG. **1**B. The lens structure **243** is configured to allow another one of the at least two optical signals emitted from the photonic component **221** to pass through. In addition, the optical guiding element **25** may be disposed on the substrate **20** and substantially aligned with the lens structure **243**. In some embodiments of the present disclosure, the optical guiding element **25** includes a reflective element. That is, the optical guiding element **25** may be used to change the direction of an incoming light beam.

[0041] The housing **21** may be disposed on the substrate and cover the photonic component **221**, the electronic component **223**, the photonic component **231**, the electronic component **232**, the laser component **233**, the lens structures **241**, **242** and **243** and the optical guiding element **25**. The housing **21** is the same as, or similar to, the housing **11** shown in FIG. **1**A and FIG. **1**B. That is, the housing **21** is configured to protect the optical transceiver module **22** and the optical emission module **23**. The housing **21** may include an opening **211** at its lateral side and an opening **212** at its upper side. As shown in FIG. **2**, the opening **211** may be substantially aligned with the lens structure **242** in a horizontal direction, and the opening **212** may be substantially aligned with the optical guiding element **25** in a vertical direction.

[0042] Referring to FIG. **2**, the optical emission module **23** with the photonic component **231**, the electronic component **232** and the laser component **233** may emit an optical signal L**1**′ to the optical transceiver module **22** with the photonic component **221** and the electronic component **223**. The optical signal L**1**′ may pass through the lens structure **241**. After the optical transceiver module **22** receives the optical signal L**1**′ emitted from the optical emission module **23**, the optical transceiver module **22** may emit optical signals L**2**′ and L**3**′. The optical signal L**2**′ may pass through the lens structure **242** and advance toward the opening **211** of the housing, thereby passing through the opening **211** of the housing **21** and emitting towards an exterior of the housing **21** of the optoelectronic structure **2**. The optical signal L**3**′ may pass through the lens structure **243** and be redirected towards the opening **212** by the reflection from the optical guiding element **25**, thereby passing through the opening **212** of the housing **21** and emitting towards an exterior of the housing **21** of the optoelectronic structure **2**.

[0043] As shown in FIG. **2**, a path of the optical signal L**3**′ may overlap a path of the optical signal L**2**′ and/or a path of the optical signal L**1**′ in a direction Y**2**, which may be substantially perpendicular to the path of the optical signal L**1**′, L**2**′ or L**3**′. Further, the path of the optical signal L**3**′ may be substantially parallel to the path of the optical signal L**1**′ and/or the path of the optical signal L**2**′. Moreover, as abovementioned, the optical signal L**3**′ may be redirected towards the opening **212** by the reflection from the optical guiding element **25**, and thus a path of the optical signal L**3**′ toward the opening **212** may be substantially perpendicular to the path of the optical signal L**3**′ toward the optical guiding element **25**.

[0044] FIG. **3** is a schematic cross-sectional view of an optoelectronic structure **3** in accordance with an embodiment of the instant disclosure. As shown in FIG. **3**, the optoelectronic structure **3** may include a substrate **30**, a photonic component **321**, an electronic component **323**, a photonic component **331**, an electronic component **332** and a laser component **333**. The substrate **30** may be or include a carrier. The substrate **30** may be composed of a silicon-based material, leveraging the well-established properties of silicon to enhance the device's performance and integration capabilities. Silicon-based substrates are chosen for their excellent electrical, thermal, and mechanical properties, making them ideal for a wide range of applications. The substrate **30** may include an interconnection structure, which may include such as a plurality of conductive traces and/or a plurality of conductive vias. The interconnection structure may include a redistribution layer (RDL) **303**. The redistribution layer **303** may be disposed on an upper surface **301** of the substrate **30**.

[0045] Referring to FIG. **3**, the photonic component **321** may be disposed over the substrate **30** and the electronic component **323** may be stacked on the photonic component **321**. In some embodiments of the present disclosure, the photonic component **321** includes photonic integrated circuits (PIC). The photonic integrated circuits may be used for emitting laser pulses, guiding the optical path, and receiving the light reflected back. Moreover, the PIC of the photonic component **321** may include a waveguide (not shown). The waveguide is configured to direct light waves from one point to another within the chip. By precisely controlling the shape and material properties of the waveguides, light's propagation path can be effectively managed, facilitating functions such as routing, splitting, combining, and modulating optical signals. As shown in FIG. **3**, the photonic component **321** may include a surface **3211** (e.g., a lower surface) facing the substrate **30** and a surface **3210** (e.g., an upper surface) opposite to the surface **3211**. Further, the surface **3211** may include an active surface of the photonic component **321**. A plurality of electrical connections **3213** may be disposed between the surface **3211** of the photonic component **321** and the upper surface **301** of the substrate **30** and configured to electrically connect the active surface of the photonic component **321** to the redistribution layer **303** of the substrate **30**. That is, the photonic component **321** and the substrate **30** may be electrically connected to each other. In some embodiments of the present disclosure, the electrical connection **3213** includes a micro-bump. Moreover, the photonic component **321** may include a plurality of conductive vias **3215** passing through it. The conductive vias **3215** may be electrically connected to the electrical connections **3213**. In some embodiments of the present disclosure, the conductive via **3215** includes a through silicon via (TSV).

[0046] In some embodiments of the present disclosure, the electronic component **323** includes electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **3**, the electronic component **323** may include a surface **3230** (e.g., a lower surface) facing the surface **3210** of the photonic component **321** and a surface **3231** (e.g., an upper surface) opposite to the surface **3210**. Further, the surface **3230** may include an active surface of the electronic component **323**. Moreover, a plurality of electrical connections **326** is arranged between the surface **3230** of the electronic component **323** and the surface **3210** of the photonic component **321** and configured to electrically connect the active surface of the electronic component **323** to the conductive vias **3215** of the photonic component **321**. That is, the electronic component **323** and the photonic component **321** may be electrically connected to each other. In some embodiments of the present disclosure, the electrical connection **326** includes a micro-bump.

[0047] The photonic component **321** and the electronic component **323** may jointly be considered as an optical transceiver module **32**. The optical transceiver module **32** may be electrically connected to the substrate **30**.

[0048] The photonic component **331** may be disposed over the substrate **30** and the electronic component **332** and the laser component **333** may be stacked on the photonic component **331**. In some embodiments of the present disclosure, the photonic component **331** includes photonic integrated circuits (PIC). The photonic integrated circuits may be used for emitting laser pulses, guiding the optical path, and receiving the light reflected back. Moreover, the PIC of the photonic component **331** may include a waveguide (not shown). The waveguide is configured to direct light waves from one point to another within the chip. By precisely controlling the shape and material properties of the waveguides, light's propagation path can be effectively managed, facilitating functions such as routing, splitting, combining, and modulating optical signals. As shown in FIG. **3**, the photonic component **331** may include a surface **3311** (e.g., a lower surface) facing the substrate and a surface **3310** (e.g., an upper surface) opposite to the surface **3311**. Further, the surface **3311** may include an active surface of the photonic component **331**. A plurality of electrical connections **3313** may be disposed between the surface **3311** of the photonic component **331** and the upper surface **301** of the substrate **30** and configured to electrically connect the active surface of the photonic component **331** to the redistribution layer **303** of the substrate **30**. That is, the photonic component **331** and the substrate **30** may be electrically connected to each other. In some embodiments of the present disclosure, the electrical connection **3313** includes a micro-bump. Moreover, the photonic component **331** may include a plurality of conductive vias **3315** passing through it. The conductive vias **3315** may be electrically connected to the electrical connections **3313**. In some embodiments of the present disclosure, the conductive via **3315** includes a through silicon via (TSV).
[0049] In some embodiments of the present disclosure, the electronic component **332** includes electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **3**, the electronic component **332** may include a surface **3320** (e.g., a lower surface) facing the surface **3310** of the photonic component **331** and a surface **3321** (e.g., an upper surface) opposite to the surface **3320**. Further, the surface **3320** may include an active surface of the electronic component **332**. The surface **3320** of the electronic component **332** may be in abutment with the surface **3310** of the photonic component **331**, and thus the active surface of the electronic component **332** may be electrically connected to the conductive vias **3315** of the photonic component **331**.
[0050] The laser component **333** may include a distributed feedback laser. The distributed feedback laser may be used for emitting highly coherent and wavelength-precise light beams. As shown in FIG. **3**, the laser component **333** may include a surface **3330** (e.g., a lower surface) facing the surface **3310** of the photonic component **331** and a surface **3331** (e.g., an upper surface) opposite to the surface **3330**. Further, the surface **3330** may include an active surface of the laser component **333**. The surface **3330** of the laser component **333** may be in abutment with the surface **3310** of the photonic component **331**, and thus the active surface of the laser component **333** may be electrically connected to the conductive vias **3315** of the photonic component **331**.
[0051] The photonic component **331**, the electronic component **332** and the laser component **333** may jointly be considered as an optical emission module **33**. The optical emission module **33** may be electrically connected to the substrate **30**. Moreover, the optical transceiver module **32** and the optical emission module **33** may be electrically coupled to each other through the substrate **30**.
[0052] As shown in FIG. **3**, the optoelectronic structure **3** may further include lens structures **341** and **342** and an optical guiding element **35**. The lens structure **341** may be disposed between the optical transceiver module **32** and the optical emission module **33**. In some embodiments of the present disclosure, the lens structure **341** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **341** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. Moreover, the

lens structure **342** is disposed between the optical transceiver module **32** and the optical guiding element **35**. In some embodiments of the present disclosure, the lens structure **342** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **342** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. In some embodiments of the present disclosure, the optical guiding element **35** includes a beam splitter which is used to divide an incoming light beam into two parts. In some embodiments of the present disclosure, the optical guiding element **35** includes a reflective element, which is used to change the direction of an incoming light beam.

[0053] In some embodiments of the present disclosure, the optoelectronic structure **3** may further include an electronic component **36** disposed on the substrate **30**. In some embodiments of the present disclosure, the electronic component **36** includes an integrated passive device (IPD).

[0054] Further, referring to FIG. **3**, the optoelectronic structure **3** may include a housing **31**. The housing **31** may be disposed on the upper surface **301** of the substrate **30**. The housing **31** may cover the optical transceiver module **32** (which may include the photonic component **321** and the electronic component **323**), the optical emission module **33** (which may include the photonic component **331**, the electronic component **332** and the laser component **333**), the lens structures **341** and **342** and the optical guiding element **35**. That is, the housing **31** is configured to protect the optical transceiver module **32** and the optical emission module **33**. The housing **31** may include a lateral portion **314** and an upper portion **315** adjacent to the lateral portion **314**. In some embodiments of the lateral portion **314** is located at a lateral side of the housing **31**. That is, the lateral portion **314** of the housing **31** may be in abutment with the upper surface **301** of the substrate **30**. In some embodiments of the present disclosure, the upper portion **315** is located at an upper side of the housing **31**. That is, the upper portion **315** may be spaced apart from the upper surface **301** of the substrate **30**. The lateral portion **314** of the housing **31** may include an opening **311**. As shown in FIG. **3**, the opening **311** may be substantially align with the optical guiding element **35** in a direction which may be substantially parallel to the upper surface **301** of the substrate **30**. The upper portion **315** of the housing **31** may include an opening **312**. As shown in FIG. **3**, the opening **312** may be disposed directly above the optical guiding element **35** and substantially aligned with the optical guiding element **35** in a direction which may be substantially perpendicular to the upper surface **301** of the substrate **30**.

[0055] As shown in FIG. **3**, the housing **31** may include a heat sink **310** at its upper side. In some embodiments of the present disclosure, the housing **31** may include a thermal conductive material. The electronic component **323** may be connected to the upper side of the housing **31**. The surface **3231** of the electronic component **323** may be attached to the upper side of the housing **31** through a thermal interface material (TIM) **327**. In some embodiments of the present disclosure, the thermal interface material **327** includes a thermal paste. That is, the housing **31** with the heat sink **310** and the thermal interface material **327** may provide a thermal dissipation path for the optical transceiver module **32**. Moreover, the electronic component **332** and the laser component **333** may be connected to the upper side of the housing **31**. The surface **3321** of the electronic component **332** and the surface **3331** of the laser component **333** may be attached to the upper side of the housing **31** through a thermal interface material **337**. In some embodiments of the present disclosure, the thermal interface material **337** includes a thermal paste. That is, the housing **31** with the heat sink **310** and the thermal interface material **337** may provide a thermal dissipation path for the optical emission module **33**. In addition, because the upper surfaces of the electronic component **323**, the electronic component **332** and the laser component **333** may be all attached to the upper side of the housing **31**, it can be understood that the upper surfaces of the electronic component **323**, the electronic component **332** and the laser component **333** may be substantially coplanar with each other.

[0056] FIG. **4** is a schematic cross-sectional view of an optoelectronic structure **4** in accordance with an embodiment of the instant disclosure. As shown in FIG. **4**, the optoelectronic structure **4** may include a substrate **40**, a photonic component **421**, an electronic component **423**, an electronic component **432** and a laser component **433**. The substrate **40** may be or include a carrier. The substrate **40** may be composed of a silicon-based material, leveraging the well-established properties of silicon to enhance the device's performance and integration capabilities. Silicon-based substrates are chosen for their excellent electrical, thermal, and mechanical properties, making them ideal for a wide range of applications. The substrate **40** may include an interconnection structure, which may include such as a plurality of conductive traces and/or a plurality of conductive vias. The interconnection structure may include a redistribution layer (RDL) **403**. The redistribution layer **403** may be disposed on an upper surface **401** of the substrate **40**.

[0057] Referring to FIG. **4**, the photonic component **421** may be disposed over the substrate **40** and the electronic component **423**, the electronic component **432** and the laser component **433** may be stacked on the photonic component **421**. In some embodiments of the present disclosure, the photonic component **421** includes photonic integrated circuits (PIC). The photonic integrated circuits may be used for emitting laser pulses, guiding the optical path, and receiving the light reflected back. Moreover, the PIC of the photonic component **421** may include a waveguide **4215**. The waveguide **4215** is configured to direct light waves from one point to another within the chip. By precisely controlling the shape and material properties of the waveguides, light's propagation path can be effectively managed, facilitating functions such as routing, splitting, combining, and modulating optical signals. In some embodiments of the present disclosure, the photonic component **421** may integrates the photonic component **121** and the photonic component **131** as shown in FIG. **1**A and FIG. **1**B. That is, the photonic component **421** may include the photonic component **121** and the photonic component **131**. As shown in FIG. **4**, the photonic component **421** may include a surface **4210** (e.g., an upper surface) facing away from the substrate **10**. Further, the surface **4210** may include an active surface of the photonic component **421**. Moreover, an electrical connection **425** may electrically connect the surface **4210** (with the active surface) of the photonic component **421** to the redistribution layer **403** of the substrate **40**. Thus, the photonic component **421** may be electrically connected to the substrate **40** through the electrical connection **425**. In some embodiments of the present disclosure, the electrical connection **425** includes a conductive wire. Since the photonic component **421** may integrates may integrate at least two photonic components, such as the photonic component **121** and the photonic component **131** as shown in FIG. **1**A and FIG. **1**B, the risk of optical misalignment in the assembly of optoelectronic structure **4** could be reduced.

[0058] The electronic component **423** may include electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **4**, the electronic component **123** may include a surface **4230** (e.g., a lower surface) facing the surface **4210** of the photonic component **421** and a surface **4231** (e.g., a lower surface) opposite to the surface **4230**. Further, the surface **4230** may include an active surface of the electronic component **423**. That is, the active surface of the photonic component **421** may face the active surface of the electronic component **423**. Moreover, a plurality of electrical connections **426** may be arranged between the surface **4210** of the photonic component **421** and the surface **4230** of the electronic component **423** and configured to electrically connect the active surface of the electronic component **423** to the active surface of the photonic component **421**. Thus, the photonic component **421** and the electronic component **423** may be electrically connected to each other through the electrical connections **426**. In some embodiments of the present disclosure, the electrical connection **426** includes a micro-bump.

[0059] The electronic component **432** may include electronic integrated circuits (EIC). The electronic integrated circuits may be used for processing signals from photodetectors, including signal amplification, filtering, analog-to-digital conversion (converting analog signals to digital signals), and performing initial data processing. As shown in FIG. **4**, the electronic component **423** may include a surface **4320** (e.g., a lower surface) facing the surface **4210** of the photonic component **421** and a surface **4321** (e.g., an upper surface) opposite to the surface **4320**. Further, the surface **4320** may include an active surface of the electronic component **432**. That is, the active surface of the photonic component **421** may face the active surface of the electronic component **432**. The surface **4210** of the photonic component **421** may be in abutment with the surface **4320** of the electronic component **432**. Thus, the photonic component **421** and the electronic component **432** may be configured in a chip-to-chip assembly and electrically connected to each other.

[0060] The laser component **433** may include a distributed feedback laser. The distributed feedback laser may be used for emitting highly coherent and wavelength-precise light beams. As shown in FIG. **4**, the laser component **133** may include a surface **4330** (e.g., a lower surface) facing the surface **4210** of the photonic component **421** and a surface **4331**(e.g., an upper surface) opposite to the surface **4330**. Further, the surface **4330** may include an active surface of the laser component **433**. That is, the active surface of the photonic component **421** may face the active surface of the laser component **433**. The surface **4210** of the photonic component **421** may be in abutment with the surface **4330** of the laser component **433**. Thus, the photonic component **421** and the laser component **433** may be configured in a chip-to-chip assembly and electrically connected to each other.

[0061] As shown in FIG. **4**, the optoelectronic structure **4** may further include lens structure **442** and an optical guiding element **45**. The lens structure **442** is disposed between the optical transceiver module **42** and the optical guiding element **45**. In some embodiments of the present disclosure, the lens structure **442** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **442** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. In some embodiments of the present disclosure, the optical guiding element **45** includes a beam splitter which is used to divide an incoming light beam into two parts. In some embodiments of the present disclosure, the optical guiding element **45** includes a reflective element, which is used to change the direction of an incoming light beam.

[0062] In some embodiments of the present disclosure, the optoelectronic structure **4** may further include an electronic component **46** disposed on the substrate **30**. In some embodiments of the present disclosure, the electronic component **46** includes an integrated passive device (IPD).

[0063] Further, referring to FIG. **4**, the optoelectronic structure **4** may include a housing **41**. The housing **41** may be disposed on the upper surface **401** of the substrate **40**. The housing **41** may cover the photonic component **421**, the electronic component **423**, the electronic component **432** and the laser component **433**, the lens structure **442** and the optical guiding element **45**. That is, the housing **41** is configured to protect the optical transceiver module **42** and the optical emission module **43**. The housing **41** may include a lateral portion **414** and an upper portion **415** adjacent to the lateral portion **414**. In some embodiments of the lateral portion **414** is located at a lateral side of the housing **41**. That is, the lateral portion **414** of the housing **41** may be in abutment with the upper surface **401** of the substrate **40**. In some embodiments of the present disclosure, the upper portion **415** is located at an upper side of the housing **41**. That is, the upper portion **415** may be spaced apart from the upper surface **401** of the substrate **40**. The lateral portion **414** of the housing **41** may include an opening **411**. As shown in FIG. **4**, the opening **411** may be substantially align with the optical guiding element **45** in a direction which may be substantially parallel to the upper surface **401** of the substrate **40**. The upper portion **415** of the housing **41** may include an opening **412**. As shown in FIG. **4**, the opening **412** may be disposed directly above the optical guiding element **45** and substantially aligned with the optical guiding element **45** in a direction which may be substantially perpendicular to the upper surface **401** of the substrate **40**.

[0064] As shown in FIG. **4**, the housing **41** may include a heat sink **410** at its upper side. In some embodiments of the present disclosure, the housing **41** may include a thermal conductive material.

The electronic component **423** may be connected to the upper side of the housing **41**. The surface **4231** of the electronic component **423** may be attached to the upper side of the housing **41** through a thermal interface material (TIM) **427**. In some embodiments of the present disclosure, the thermal interface material **427** includes a thermal paste. That is, the housing **41** with the heat sink **410** and the thermal interface material **427** may provide a thermal dissipation path for the optical transceiver module **42**. Moreover, the electronic component **432** and the laser component **433** may be connected to the upper side of the housing **41**. The surface **4321** of the electronic component **432** and the surface **4331** of the laser component **433** may be attached to the upper side of the housing **41** through a thermal interface material (TIM) **437**. In some embodiments of the present disclosure, the thermal interface material includes thermal paste. That is, the housing **41** with the heat sink **410** and the thermal interface material **437** may provide a thermal dissipation path for the optical emission module **43**. In addition, because the upper surfaces of the electronic component **423**, the electronic component **432** and the laser component **433** may be all attached to the upper side of the housing **41**, it can be understood that the upper surfaces of the electronic component **423**, the electronic component **432** and the laser component **433** may be substantially coplanar with each other.

[0065] FIG. **5**A, FIG. **5**B, FIG. **5**C, FIG. **5**D, FIG. **5**E, FIG. **5**F, FIG. **5**G, FIG. **5**H, FIG. **5**I, FIG. **5**J and FIG. **5**K illustrate one or more stages of an example of a method for manufacturing an optoelectronic structure in accordance with some embodiments of the present disclosure.

[0066] Referring to FIG. **5**A, a photonic component **521** is provided. In some embodiments of the present disclosure, the photonic component **521** includes photonic integrated circuits (PIC).

[0067] Referring to FIG. **5**B, an electronic component **523** is provided. In some embodiments of the present disclosure, the electronic component **523** includes electronic integrated circuits (EIC). The electronic component **523** may be mounted on the photonic component **521** and electrically connected to the photonic component **521** through electrical connections **526**.

[0068] The photonic component **521** and the electronic component **523** may jointly be an optical transceiver module **52**.

[0069] Referring to FIG. **5**C, a photonic component **531** is provided. In some embodiments of the present disclosure, the photonic component **531** includes photonic integrated circuits (PIC).

[0070] Referring to FIG. **5**D, an electronic component **532** is provided. In some embodiments of the present disclosure, the electronic component **532** includes electronic integrated circuits (EIC). The electronic component **532** may be mounted on the photonic component **531**. The photonic component **531** and the electronic component **532** may be configured in a chip-to-chip assembly and electrically connected to each other.

[0071] Referring to FIG. **5**E, a laser component **533** is provided. In some embodiments of the present disclosure, the laser component **533** includes a distributed feedback laser. The laser component **533** may be mounted on the photonic component **531**. The photonic component **531** and the laser component **533** may be configured in a chip-to-chip assembly and electrically connected to each other.

[0072] The photonic component **531**, the electronic component **532** and the laser component **533** may jointly be an optical emission module **53**.

[0073] Referring to FIG. **5**F, a substrate **50** is provided. The substrate **50** may be composed of a silicon-based material. The substrate **50** may include a redistribution layer **503**.

[0074] Referring to FIG. **5**G, the optical transceiver module **52** and the optical emission module **53** are attached to an upper surface **501** of the substrate **50**. Moreover, an electronic component **56** is mounted to the substrate **50**. In some embodiments of the present disclosure, the electronic component **56** includes an integrated passive device (IPD).

[0075] Referring to FIG. **5**H, electrical connections **525** and **535** are provided. The electrical connection **525** may connect the photonic component **521** of the optical transceiver module **52** to the substrate **50** and serve to establish electrical connectivity between the optical transceiver module **52** and the substrate **50**. The electrical connection **535** may connect the photonic component **531** of the optical emission module **53** to the substrate **50** and serve to establish electrical connectivity between the optical emission module **53** and the substrate **50**.

[0076] Referring to FIG. **5**I, lens structures **541** and **542** and an optical guiding element **55** are provided. The lens structures **541** and **542** and an optical guiding element **55** may be attached to the upper surface **501** of the substrate **50**. In some embodiments of the present disclosure, the lens structure **541**, **542** includes a plurality of lenses. In some embodiments of the present disclosure, the lens structure **541**, **542** is configured to collimate the optical signal from a divergent light into a collimated or parallel light beam having a consistent beam size. In some embodiments of the present disclosure, the optical guiding element **55** includes a beam splitter which is used to divide an incoming light beam into two parts. In some embodiments of the present disclosure, the optical guiding element **55** includes a reflective element, which is used to change the direction of an incoming light beam.

[0077] Referring to FIG. **5**J, a housing **51** is provided. The housing **51** may be attached to the upper surface **501** of the substrate **50**. The housing **51** may include an opening **511**, which is located at a lateral side of the housing **51** and substantially aligned with the optical guiding element **55** in a horizontal direction, and an opening **512**, which is located at an upper side of the housing **51** and substantially aligned with the optical guiding element **55** in a vertical direction. Moreover, the electronic component **523** may be attached to the housing **51** through the thermal interface material **527**, and the electronic component **532** and the laser component **533** may be attached to the housing **51** through the thermal interface material **537**.

[0078] Referring to FIG. **5**K, a singulation process is performed by cutting through the substrate **50**. The singulation may be performed, for example, by using a dicing saw, laser or other appropriate cutting technique.

[0079] After the manufacturing process as shown in FIG. **5**A, FIG. **5**B, FIG. **5**C, FIG. **5**D, FIG. **5**E, FIG. **5**F, FIG. **5**G, FIG. **5**H, FIG. **5**I, FIG. **5**J and FIG. **5**K, the optoelectronic structure **5** is formed (see FIG. **5**K). In some embodiments of the present disclosure, the optoelectronic structure **5** is the same as, or similar to, the optoelectronic structure **1** shown in FIG. **1**A and FIG. **1**B.

[0080] The present disclosure relates an optoelectronic structure that stacks an electronic component with an Electronic Integrated Circuit (EIC) and a photonic component with a Photonic Integrated Circuit (PIC). This configuration forms an optical transceiver module and an optical emission module, which are then assembled separately onto a carrier. This assembly method effectively minimizes optical alignment challenges and reduces manufacturing risks by simplifying the integration process of the optical components, thereby enhancing the efficiency and reliability of the optoelectronic structure in applications requiring precise optical and electronic integration.

[0081] Furthermore, the optoelectronic structure may accommodate optical elements like lenses and reflective mirrors within its housing, enabling the emission of at least two light beams in distinct directions. This capability allows the structure to perform LiDAR scanning across different orientations, enhancing its ability to detect and measure the distance to objects in its environment with greater accuracy and efficiency. The inclusion of such optical components within a single housing simplifies the system's design while expanding its functional versatility in LiDAR applications.

[0082] As used herein, the singular terms “a,” “an,” and “the” may include a plurality of referents unless the context clearly dictates otherwise.

[0083] As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation of less than or equal to ±10% of that numerical value, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. For example, two numerical values can be deemed to be "substantially" the same or equal if the difference between the values is less than or equal to ±10% of an average of the values, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. For example, "substantially" parallel can refer to a range of angular variation relative to 0° that is less than or equal to ±10°, such as less than or equal to ±5°, less than or equal to ±4°, less than or equal to ±3°, less than or equal to ±2°, less than or equal to ±1, less than or equal to ±0.5°, less than or equal to ±0.1°, or less than or equal to ±0.05°. For example, "substantially" perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to ±10°, such as less than or equal to ±5°, less than or equal to ±4°, less than or equal to ±3°, less than or equal to ±2°, less than or equal to ±1, less than or equal to ±0.5°, less than or equal to ±0.1°, or less than or equal to ±0.05°.

[0084] Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly specified.

[0085] While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations do not limit the present disclosure. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not be necessarily drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein are described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations on the present disclosure.

Claims

1. An optoelectronic structure, comprising: an optical device; and a housing covering the optical device; wherein the optical device is configured to transmit at least one optical signal to an outside of the housing in different directions.

2. The optoelectronic structure of claim 1, wherein the at least one optical signals comprises a first optical signal and a second optical signal, wherein the housing comprises a first opening and a second opening, and wherein the first opening is located at an upper portion of the housing and configured to allow the first optical signal to pass through and the second opening is located at lateral portion of the housing and configured to allow the second signal to pass through.

3. The optoelectronic structure of claim 2, further comprising an optical element covered by the housing and disposed below the first opening, and wherein the optical element is configured to alter a path of the at least one optical signal.

4. The optoelectronic structure of claim 3, further comprising a lens structure disposed between the optical element and the optical device.

5. The optoelectronic structure of claim 1, further comprising a carrier supporting the housing and electrically connected to the optical device.

6. The optoelectronic structure of claim 1, wherein the optical device is configured to receive an optical signal reflected from an external object external to the optoelectronic structure.

7. An optoelectronic structure, comprising: an optical emission module; and an optical transceiver module configured to optically couple to the optical emission module and configured to emit a first optical signal along a first path adjacent the optical transceiver module.

8. The optoelectronic structure of claim 7, wherein the first optical signal is emitted in a first direction, the optical emission module is configured to emit a second optical signal in a second direction opposite to the first direction.

9. The optoelectronic structure of claim 8, wherein the optical transceiver module is configured to receive the second optical signal from a lateral side of the optical transceiver module and is configured to emit the second optical signal from the lateral side.

10. The optoelectronic structure of claim 8, further comprising a first optical element configured to guide the first optical signal into a first beam emitted in a third direction and a second beam in a fourth direction different from the third direction.

11. The optoelectronic structure of claim 10, wherein the first direction is substantially equal to the third direction.

12. The optoelectronic structure of claim 10, wherein the third direction and the fourth direction are substantially orthogonal to each other.

13. An optoelectronic structure, comprising: a housing having a first opening and a second opening; and an optical device covered by the housing; wherein the optical device is configured to emit and/or receive a light passing through the first opening of the housing and configured to emit and/or receive another light passing through the second opening of the housing.

14. The optoelectronic structure of claim 13, wherein the housing has a first side portion and a second side portion different from the first side portion, and wherein the first opening is located at the first side portion and the second opening is located at the second side portion.

15. The optoelectronic structure of claim 14, wherein the first side portion is adjacent to the second side portion.

16. The optoelectronic structure of claim 14, further comprising a carrier disposed below the housing, wherein the first side portion of the housing connects an upper surface of the carrier and the second side portion of the housing is spaced apart from the upper surface of the housing.

17. The optoelectronic structure of claim 16, wherein the optical device comprises an optical emission module and an optical transceiver module disposed over the carrier respectively, and wherein the optical emission module and the optical transceiver module collectively supports the housing.

18. The optoelectronic structure of claim 17, wherein the optical transceiver module comprises a photonic component connected to the carrier and an electronic component disposed over the photonic component, and wherein the electronic component is electrically connected to the carrier through the photonic component.

19. The optoelectronic structure of claim 13, further comprising a first optical element disposed between the housing and the optical device, wherein the first optical element is configured to guide a first light beam toward the first opening.

20. The optoelectronic structure of claim 19, further comprising a second optical element disposed between the housing and the optical device, wherein the second optical element is configured to guide a second light beam toward the second opening.